

GeoGuard

Satellite Air Quality Intelligence

Dated atmospheric measurements and AI-assisted methane screening for environmental investigation

Team GeoGuard · Theme: Air Quality Intelligence · UAE pilot

Environmental investigation

User

**Environmental monitoring
and enforcement teams**

Decision

**Where should we review
satellite evidence and
request field measurements?**

The problem

Separate products, mixed units and missing observations complicate comparisons. Satellite coverage provides regional context; dated evidence supports closer review.

Measurements in the PoC

Quantity	Sentinel-5P measurement	Display
NO ₂	Tropospheric column	μmol/m ²
CO	Total atmospheric column	mmol/m ²
SO ₂	Vertical column; assumed ground profile	μmol/m ²
Regional CH ₄	Bias-corrected dry-air column mixing ratio	ppb

Sentinel-2 imagery + MARS-S2L: candidate / no candidate / not assessable

PM, VOC and H₂S instruments are not connected. Columns are not surface exposure.

Satellite evidence workflow

Atmospheric measurements



Methane screening



Different quantities and footprints remain separate. Missing observations stay unknown.

UAE atmospheric comparison

Dubai and Jebel Ali · 5 km radius comparison circles

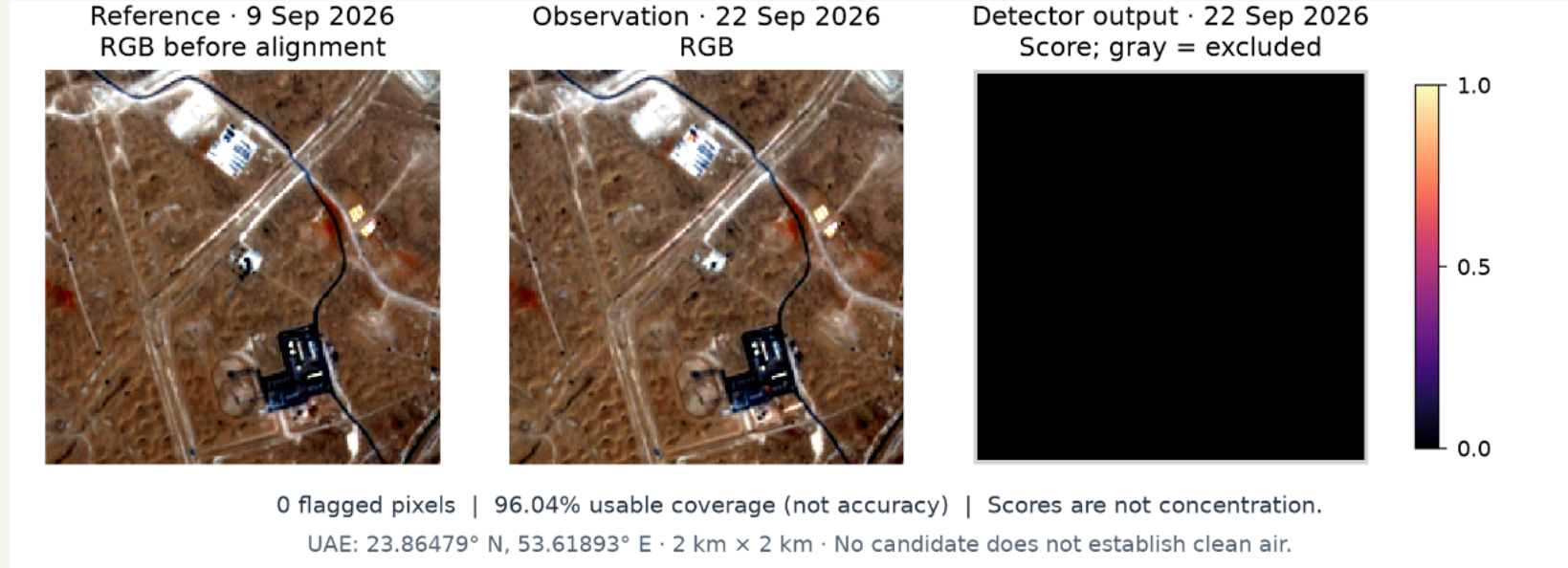
Quantity / window	Dubai mean · usable days	Jebel Ali mean · usable days
NO ₂ · Jan–Aug 2026	207.85 $\mu\text{mol}/\text{m}^2$ · 226/243	206.67 $\mu\text{mol}/\text{m}^2$ · 224/243
CO · Aug–Sep 2026	36.33 mmol/m^2 · 52/61	37.70 mmol/m^2 · 60/61
SO ₂ · Aug–Sep 2026	145.83 $\mu\text{mol}/\text{m}^2$ · 60/61	143.96 $\mu\text{mol}/\text{m}^2$ · 60/61
Regional CH ₄ · Aug–Sep 2026	1966.93 ppb · 33/61	1969.56 ppb · 37/61

Different valid days and retrieval uncertainty limit comparisons.

These means alone do not establish significance or identify an emission source.

Dated methane screening

22 September 2026 observation · 9 September reference
23.86479° N, 53.61893° E



0

candidate pixels

96.04%

usable coverage

No candidate is not proof of zero methane. Coverage is not model accuracy.

Review an area in GeoGuard

1 Choose an area

Select a UAE location and quantity.

2 Read the evidence

Review units, observation dates, coverage and history.

3 Inspect methane candidates

Review dated images and quality information.

4 Export and follow up

Share evidence and request gas-specific field checks.

Bundled NO₂ has area/history results; CO, SO₂ and regional CH₄ include archived map layers.

Reproducibility and validation

30 software tests passed

The PoC runs end to end with pinned dependencies and bundled real inputs. Daily, monthly and period summaries match the archived Earth Engine exports.

Executed notebook, raw inputs, numerical outputs and evidence are available for review. Website checks cover quantity switching, imports and location matching.

Software verification does not establish independent UAE detection accuracy.

Interpretation and development priorities

Current limits

Columns do not give ground-level exposure, AQI, emission rates or a safety verdict.

Candidates need analyst review. Clouds, wind and reference uncertainty matter.

UAE forecasting is not validated.

Next evidence needed

Pair satellite observations with calibrated, gas-specific field inputs.

Build reviewed plume/no-plume history and test on held-out dates and locations.

Use persistent hosting for unattended checks.

The PoC supports environmental investigation; it does not confirm emissions.

Team and demonstration

Zainab

Pollutants, sources
and sensing methods

Raseel

Earth Engine mapping
and comparison areas

Abdulaziz

AI methane screening
and project integration

Website

losth6wz.github.io/Geoguard/

Repository and reproducible PoC

github.com/losth6wz/Geoguard

Data and model attribution

Copernicus Sentinel-5P / Sentinel-2

2026 satellite observations; Earth Engine and public imagery archives

UNEP IMEO MARS-S2L + CloudSEN12

Pretrained models; MARS-S2L CC BY-NC-SA 4.0, CloudSEN12 weights CC BY-NC 4.0

NASA GEOS-FP / recorded GFS fallback

Modeled wind matched to acquisition time; source and fallback recorded

Full references and applicable terms: `THIRD_PARTY_NOTICES.md` in the repository